RedBus Smart Travel Assistant Chatbot



Overview

What is Redbus?

RedBus is India's largest online bus ticketing platform, founded in 2006 and now a part of the MakeMyTrip Group . It provides intercity bus ticket bookings via website, mobile app, and APIs. It connects travelers with a wide network of bus operators across India and other countries like Singapore, Malaysia, Indonesia, and Colombia.

Value Proposition

- Passengers: Seamless online booking, operator reviews, price comparison, real-time tracking, customer support
- Bus Operators: Access to a large digital user base, tech platform for fleet management, ticket distribution
- Partners: APIs for integration, additional distribution channels, commission-based revenue

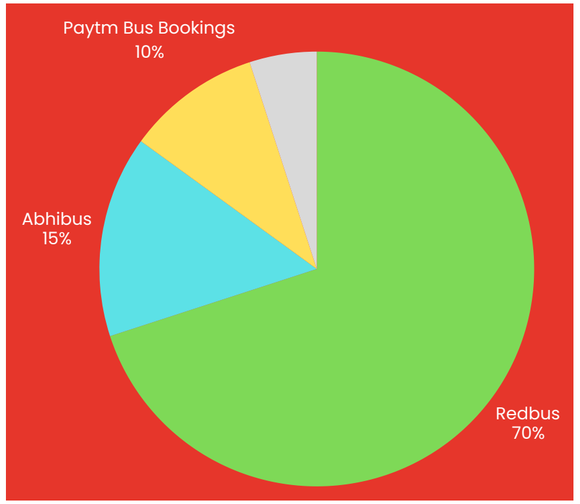
Revenue Model

- Ticket Commission: RedBus earns 10–15% on each ticket sold through the platform.
- Advertising and Promotions: Bus operators and partners can pay to promote routes, appear higher in listings, or run offers.
- Value-Added Services: Revenue from services like travel insurance, seat selection, cancellation protection.
- B2B Solutions (BOSS Platform): Provides SaaS tools for fleet operators to manage bookings, schedules, and CRM.

Actors

Passengers	Book tickets, rate experiences
Bus Operators	List and manage trips, receive bookings
Affiliates & OTAs	Distribute tickets through APIs
RedBus Tech Team	Platform development, AI, routing algorith
Customer Support	Assist with disputes, cancellations, refunds
Payment Partners	Enable secure online transactions

Market Share



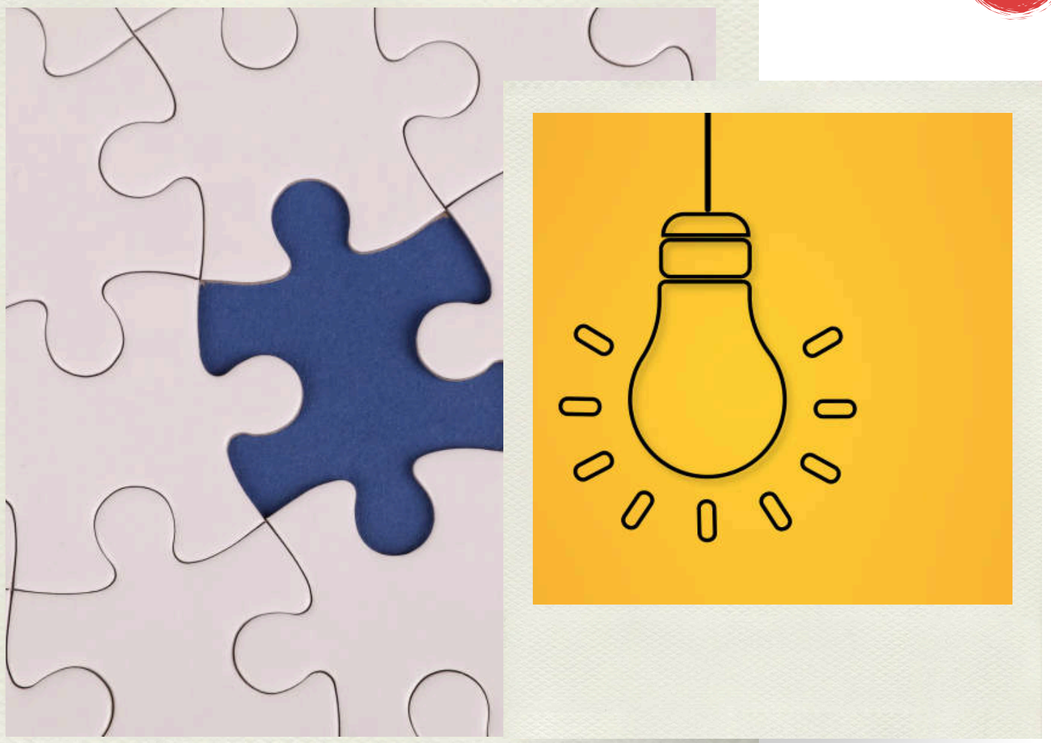
Travel Assistant- Jobs to be Done

- Problem Statement: Users are unsure of optimal routes, especially in unfamiliar cities or during multi-leg journeys.Sometimes the exact destination is not served directly by buses.They don’t want to browse dozens of buses and routes manually.

User Goal	What the Chatbot Helps With
I want to book a bus to a location that doesn't show up	Suggests **nearest reachable locations** and guides on how to reach the actual destination
I need the **fastest route** to my destination	Filters buses based on **shortest total travel time**, including multi-hop routes
I want the **cheapest way** to travel	Recommends **cost-optimized options**, even if they involve combining bus + local transport
I don’t know the **exact name** of the place	Uses **fuzzy location matching** to detect and suggest the correct or nearby destination
I missed my bus or plan changed last minute	Shows **alternate route options** from current or nearby locations quickly
I want to reach a **remote town or village**	Offers **intelligent break-journey combinations** using local transport knowledge
I want suggestions **without browsing 100s of options**	Provides **personalized bus suggestions** directly in the chat with booking buttons
I want to feel **confident about the route** I’m taking	Gives **route rationale**, map previews, estimated timings, and mode-switch suggestions







Solution

To enhance the chatbot's intelligence by enabling it to:

- Suggest the best route combinations to reach the destination
- Recommend nearest boarding/dropping points if the destination is unavailable
- Provide smart filters like fastest, cheapest, most comfortable
- Offer multi-hop suggestions when a direct bus isn't available

User Flow

User: I want to go to Kudremukh

Bot: "We don't have direct buses to Kudremukh. But I found the best way to reach there!"

➡ Option 1:

- Bus: Bangalore → Chikmagalur
- Local cab: Chikmagalur → Kudremukh (40 km)
- Total: 9 hours | ₹850

➡ Option 2:

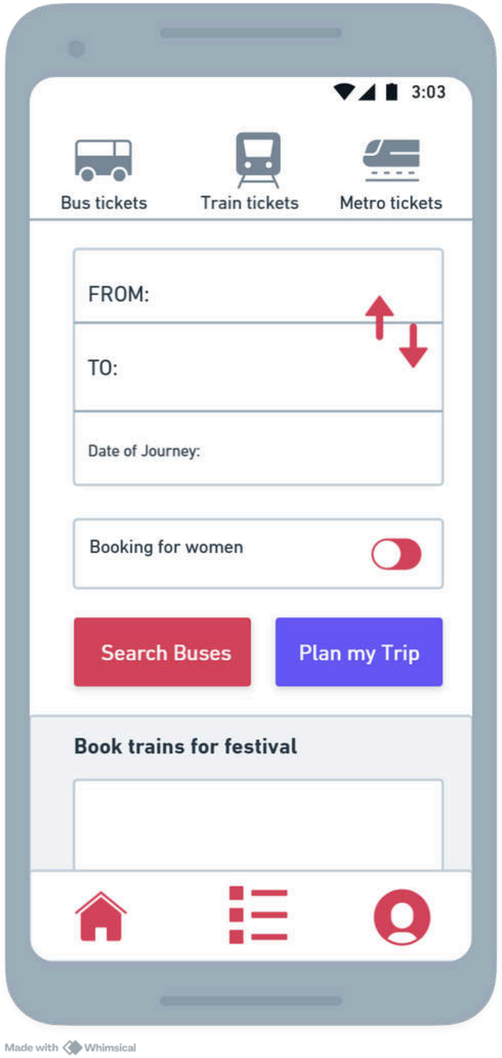
- Bus: Bangalore → Kalasa
- Local Jeep: Kalasa → Kudremukh
- Total: 8.5 hours | ₹820

[Choose Option 1], [Choose Option 2], [Edit my destination]

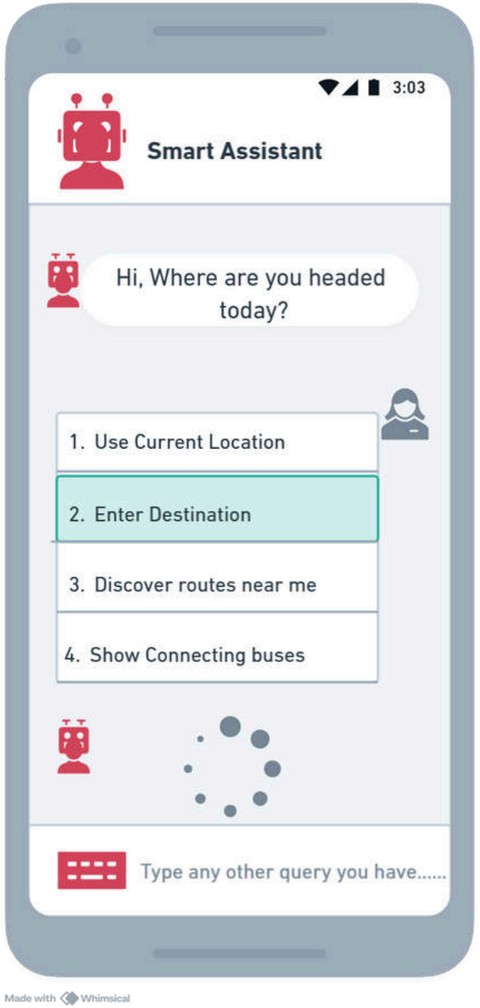
Key AI Components

- Location intelligence API: Google Maps / RedBus route data
- Fallback matching: Closest served bus stops via geolocation
- Multi-hop recommendation engine: Suggests partial bus + local ride combo
- Filters: Cost, duration, boarding point proximity, preferred timings
- Learning model: Learns from frequently booked alternative routes

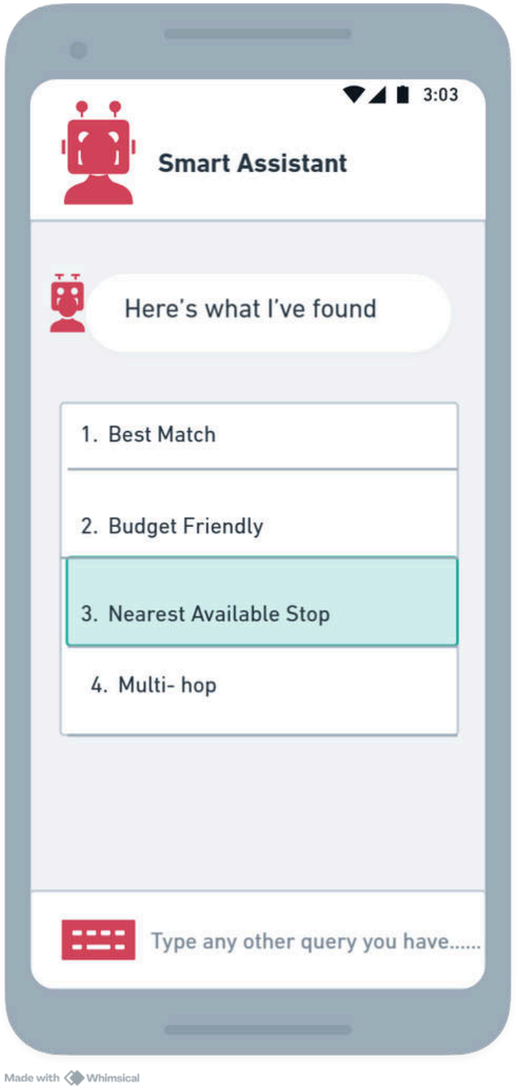
Wireframes



- Floating Chatbot Icon labeled: “Plan My Trip”
- Visible CTA near search bar: “Not sure where to go? Let’s help!”



User can type vague queries like “Need to go near Shimoga” or “Find cheapest bus tonight.” The chatbot interprets intent.



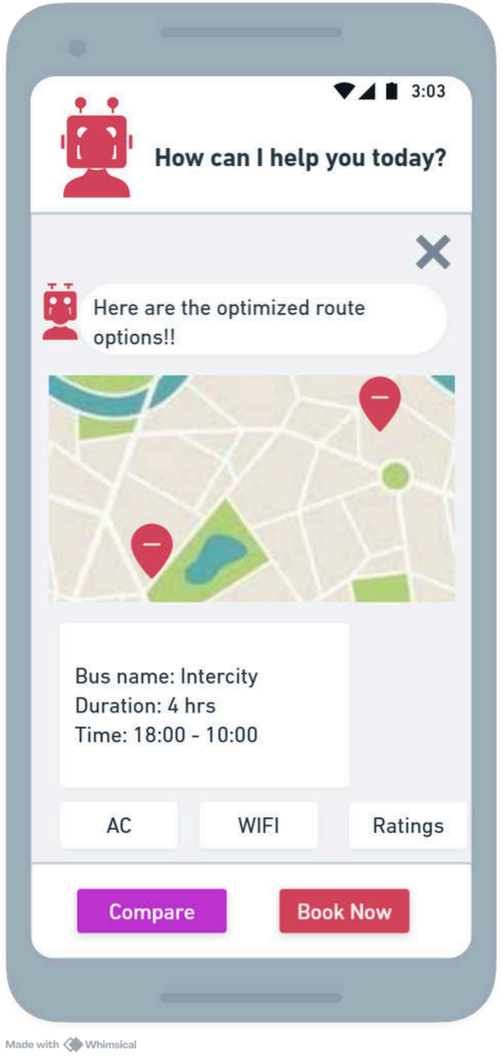
The AI ranks the routes using NLP and geolocation-based optimization, with alternatives when exact matches aren’t available.

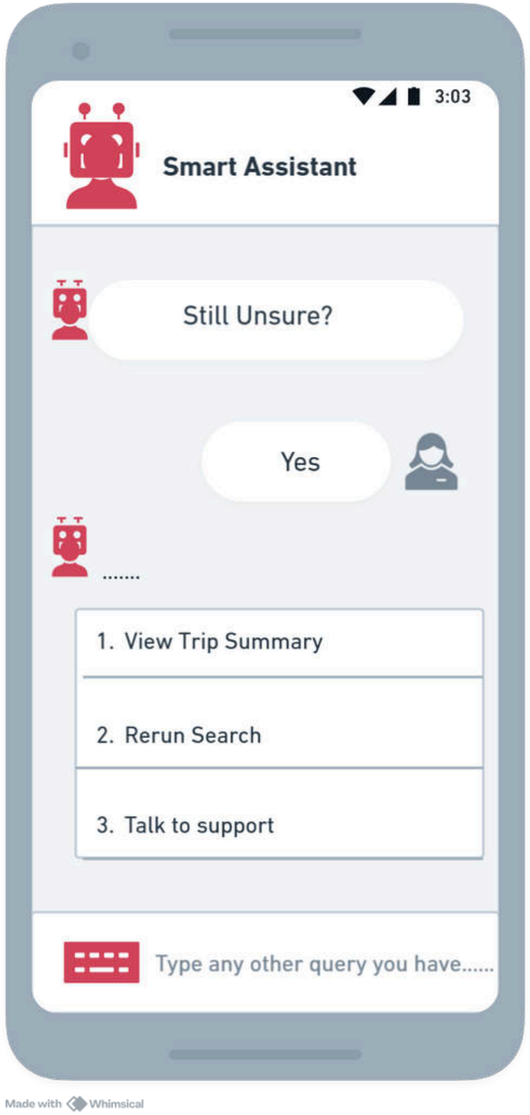
Wireframes

Route Card Details
Wireframe Elements:

- Route Preview with map snapshot
- Bus name, time, duration, fare
- Icons for WiFi, AC, Ratings
- CTA: “Book Now” / “Compare”

Flow Explanation:
Each card shows enough data for quick decision-making.
Comparison leads to modal or redirect to detail view.





Post-Selection Support
Wireframe Elements:

- “Still unsure?” prompt
- Options:
  - Talk to Support
  - Rerun Search
  - View Trip Summary

Flow Explanation:
Encourages final confirmation or iterative search. All support is conversational.

Metrics

Metric	What It Tracks
Time to First Route Recommendation	Measures how quickly the chatbot surfaces an initial route after a user query.
Chat-to-Booking Conversion Rate	Percentage of chat sessions that end in a completed ticket purchase.
Drop-Off Rate After First Interaction	Share of users who abandon the chat before receiving or selecting a suggestion.
User Satisfaction Score (CSAT)	Average rating users give the chat assistant after an interaction (1–5 scale).
Repeat Chatbot Usage	Frequency with which the same user returns to the assistant within a week.
Average Booking-Time Reduction	Improvement in total time from search start to payment confirmation vs. baseline.
Alternate Route Discoveries	Number of bookings made via AI-suggested indirect or multi-hop routes.



Thank You

